

A Tube-MPC Approach to Autonomous Multi-Vehicle Racing on High-Speed Ovals

Alexander Wischnewski , Thomas Herrmann , Frederik Werner , and Boris Lohmann 

Abstract—Autonomous vehicle racing has emerged as vibrant, innovative technology development, demonstration platform in recent years. Universities, companies demonstrate their achievements on various vehicles - from 1:10th to full-scale prototypes. One of those platforms is the Dallara AV-21, and the spec-vehicle for the Indy Autonomous Challenge. This paper outlines the robust model predictive control (MPC) concept used within the software stack of the TUM Autonomous Motorsport team. It is based on a simplified friction-limited point mass model and a set of low-level feedback controllers. The remaining model uncertainties are managed via introducing a constraint-tightening approach based on a Tube-MPC approach. In contrast to classical tracking controllers, the optimization problem is formulated to freely optimize the trajectory while staying within certain maximum deviations of the reference. This approach allows to rely on a coarse output of the trajectory planning approach while maintaining smoothness requirements in steering, throttle, and brake actuation. The paper highlights the advantages of the proposed robust reoptimization concept compared to pure tracking formulations. It showcases the performance compared to a classical LQR controller and an MPC, which utilizes a vehicle model with a more sophisticated tire model. The controller achieved a top speed of 265 kmh^{-1} and lateral accelerations up to 21 ms^{-2} during a two-vehicle competition involving dynamic overtaking maneuvers on the Las Vegas Motor Speedway, a famous racetrack with turns banked up to 20° .

Index Terms—MPC, robust, control, vehicle dynamics, autonomous driving.

I. INTRODUCTION

CONTROLLING a vehicle safely at the limits of its physical capabilities is a key requirement to enable full autonomy even in the most challenging scenarios. The main challenges from a control systems perspective are the strong coupling of longitudinal and lateral vehicle dynamics, the nonlinearities in the tire behavior, and the apparent uncertainty with respect to tire-road friction, actuator dynamics, and environmental influences. From an engineering point of view, these are accompanied



Fig. 1. The race vehicles driven by the autonomous software stacks of the TUM Autonomous Motorsport and the PoliMOVE teams at the passing competition of the Indy Autonomous Challenge at January 7th 2022 with speeds above 260 kmh^{-1} . Credits: Indy Autonomous Challenge.

by the difficulty to recreate and test scenarios that require a vehicle to operate at its limits. These are usually tried to be circumvented and therefore rare.

All of those challenges can be found in autonomous racing, which is therefore an ideal benchmark environment for control algorithms which target these problems. A team from the Technical University of Munich entered the Indy Autonomous Challenge [1], a university competition aiming at advancing the state-of-the-art in autonomous vehicle research (see Fig. 1). This paper is focused on the motion control algorithm of the software stack, which won the inaugural Indy Autonomous Challenge at the Indianapolis Motor Speedway on October 23rd 2021 and placed second in the passing competition with two vehicles at the Las Vegas Motor Speedway on January 7th 2022.

II. RELATED WORK

A. State-of-the-Art

The field of autonomous vehicle control is well established since the successful application of multiple control strategies during the DARPA autonomous driving challenges [2], [3] as well as the availability of several series-level driver assistance systems. A survey covering the key concepts used in the field is presented in [4]. In recent years, several researchers have turned towards the application of these algorithms in racing scenarios [5]–[7]. These combine the unique opportunity to benchmark algorithms close to the limit of the vehicle's capabilities while being inherently safe at the same time.

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The vehicle dynamics near the handling limits show significant nonlinearities in the tire behavior and strong coupling between the longitudinal and lateral vehicle motion. The concepts applied in [5] and [7] demonstrate successful racing on a full-size prototype using a linearized system description and adjust the linear lateral feedback controller using a gain-scheduling strategy based on the vehicle velocity. An enhancement to those approaches is the utilization of exact linearization-based controller design [8] methods; however, those perform an implicit decoupling and linearization of the multivariable control task and can therefore not fully exploit potential advantages of the nonlinear system dynamics. Furthermore, they are usually prone to parametric as well as dynamic uncertainties. A comprehensive comparison of other nonlinear control techniques which do not require online optimization can be found in [9]. All of them share the common disadvantage, that they do not explicitly handle constraint satisfaction in the controller design. This leaves the consideration of tire and position deviation constraints to an a-priori specified worst-case safety margin for the trajectory planning algorithms.

In contrast to the already mentioned approaches, MPC promises to handle the multivariable, constrained control problem via the repeated solution of a finite-horizon optimization problem during the vehicle's operation. The first systems of this type dealt with the lateral and longitudinal control problem independently [10]. However, this approach does not allow to leverage many of the strengths of MPC concepts. The majority of the approaches therefore jointly optimize the lateral and longitudinal vehicle dynamics at the expense of larger optimization problems and higher computational burden. They can be grouped according to the class of models used: Point-mass models offer a simplistic view of the dynamics while still capturing the major dynamics in the euclidean space. If they are used at the handling limits of the vehicle, they are enhanced with friction limits to account for the physical limitation of the tires [11]. More complex models usually use single-track models with varying fidelity of the tire model [6], [12], [13]. A common choice are Pacejka [14] models with a coupling via a constraint on the maximum combined tire force. The influence of the model choice on the controller capability is analyzed by [15]. It was found that the choice of vehicle model does not significantly influence the optimal racing trajectory. Still, the more complex models can consider and stabilize the transient yaw dynamics. However, at the same time, the computational complexity grows significantly due to the additional state variables and constraints which have to be considered. In addition to handling multi-input dynamics and constraints, a key strength of MPC approaches are applications where a reoptimization around an initial reference path is allowed and required [6], [16], e.g., to consider sudden obstacles or to achieve smoother input behavior. This property will also be exploited in this paper to reduce the smoothness requirements for the output of the motion planning algorithms. Even more complex vehicle dynamics descriptions, e.g., including the banking and inclination effects of racetracks, have been used for minimum lap-time optimal control problems but have not yet shown to be real-time capable [17].

The case of autonomous racing holds a unique position in these approaches as it aims at minimizing the travel time around the circuit instead of tracking a given trajectory. This requirement can be approached using minimum-time optimization [18], by reformulating the control objective to maximize the progress along a given reference track [6], [19] or via a two-level approach where an appropriate reference trajectory generation takes care of the overall aim and a high fidelity nonlinear MPC is applied to track this trajectory [6], [20].

A challenge for all of those approaches is to generate a recursively feasible formulation of the finite-horizon optimization problem. From a practical perspective, this translates into designing the optimization problem such that it applies sufficient caution to manage effects that affect the behavior after the considered optimization horizon. A theoretically rigorous solution to this problem would be the construction of a positive invariant set [21]. However, this is a yet unsolved problem in the case of arbitrary trajectory tracking with nonlinear vehicle models. Promising directions towards this aim are given via offline computation of a discriminating kernel for a given track [22] and via the usage of simplified vehicle models [23]. A way to circumvent this issue is the utilization of sufficiently long optimization horizons and the application of slack variables to soften the state constraints [18], [19].

Similar issues arise in reality due to the inevitable model inaccuracies and external disturbances. From a control-theoretic perspective, they can be resolved by thoroughly extending the recursive feasibility analysis framework to a robust version considering those uncertainties, either in a deterministic or a stochastic modeling framework. Solutions for linear models are well-known [24], [25] and extensions for nonlinear models are coming up in recent years [26]. While there already exist several applications for autonomous driving in general [27]–[29], the application to racing vehicles has not found widespread adoption so far. This is likely due to the conservative nature of thorough theoretical solutions for high-dimensional systems. Notable exceptions have been the approach proposed by [30], which models uncertainty in a stochastic framework and manages to reduce the accident rate while keeping a similar performance level, [31], which leverages constraint tightening to handle the transient dynamics of an auxiliary vehicle dynamics learning algorithm, and [32], which uses a similar approach to introduce tuning parameters for managing caution with respect to constraint exploitation in a structured way.

B. Scope and Outline of the Paper

In this paper, we extend the Tube-MPC scheme presented in [32] to match the requirements of high-speed oval racing. First, we allow for varying acceleration limits due to the large speed range (depending on the aerodynamic downforce) and propose a strategy to handle the three-dimensional layout of the race track, i.e., the effects on the vehicle dynamics due to banked turns. Second, we propose a cost design to smoothen coarse target trajectories to improve driving stability while simultaneously adhering to the specified constraints. In contrast to a classical trajectory tracking controller, this concept can handle

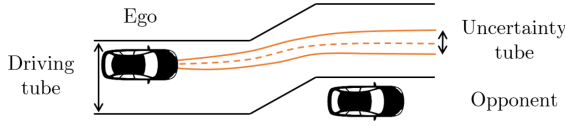


Fig. 2. Visualization of the Tube-MPC concept with reoptimization capabilities. The nominal prediction (dashed-orange) of the MPC is replaced with a set-valued prediction representing the potential uncertainty within the model (solid-orange). The controller is allowed to deviate from the middle of the driving tube but considers the constraints for all potential future outcomes of the system (black).

non-continuous curvature and longitudinal accelerations of the target trajectory without an impact on the driving performance. Third, we present results of a full-scale vehicle racing in scenarios with two vehicles (including dynamic replanning) with speeds up to 265 kmh^{-1} and accelerations of 21 ms^{-2} . Additionally, we perform a simulation study to compare the proposed friction limited point-mass model Tube-MPC with an MPC scheme based on a nonlinear tire model and a classical LQR controller and elaborate on the importance of the underlying acceleration controllers.

The paper is structured as follows: The following section presents essential preliminaries and the vehicle modeling applied in this work. Section IV explains the proposed Tube-MPC scheme. An extensive study of different algorithm configurations in simulation is presented within Section V. The successful application of the proposed algorithm on a full-scale prototype at high speeds is presented in Section VI. The conclusions drawn can be found in Section VII.

III. METHODOLOGY

A. Software Architecture

The overall driving task within the autonomous racing software stack is split into three areas: perception, planning, and control [33]. We omit a detailed discussion for the perception and the planning part and only cover the aspects that are important to understand the following implications for the control system. The perception part of the software stack builds upon multiple independent object detection pipelines using LIDAR, Radar and Camera. These are fused via an object-tracking algorithm based on a Kalman-Filter approach. Based on those estimates, a prediction algorithm forecasts what the other vehicles on track are going to do. The planning problem is difficult to be efficiently solved due to the combinatorial nature resulting from other vehicles on the track [6]. Intuitively, this reflects the fact that there are multiple local minima [34], e.g., passing a vehicle in front either on the left or on the right side. We dedicate the task of this high-level decision problem to the trajectory planning algorithm and will use a graph-based optimization framework for this (see also the discussions on this in [6], [35]). The result of this task is an admissible driving tube depicted in black in Fig. 2.

The control algorithm generates appropriate vehicle control commands in the form of throttle, brake, and steering setpoints. It can achieve its maximum potential when it is allowed to freely optimize within the admissible driving tube. In contrast to pure tracking control, this paradigm change makes it possible

to reduce the quality requirements for the trajectory planning algorithm. Instead of delivering a fine-tuned target trajectory, the planning algorithm aims to make behavioral decisions that are refined within an MPC algorithm. In addition, this allows the planner to cover longer planning horizons at the same computational budget, which is essential for high-speed operation.

The ability to exploit the admissible driving tube poses challenges when trying to guarantee that the MPC algorithm stays recursively feasible as the solutions to the optimization problem tend to be close to the constraints when the lateral deviation cost is omitted or very small. This issue arises from the fact that nominal control algorithms are prone to becoming infeasible in the presence of model uncertainties and disturbances [25]. This is countered by the explicit consideration of these in the design of the controller via the design of a Tube-MPC scheme instead of a nominal MPC. The key idea is to replace the predicted trajectory with a prediction of a set of possible trajectories resulting from the impact of uncertainties and disturbances (depicted in orange in Fig. 2). This creates a control behavior that exploits the constraints at the beginning of the optimization horizon and stays cautious towards the end.

B. Tube-MPC

We consider the discrete, linear time-invariant dynamic system

$$x(k+1) = Ax(k) + Bu(k) + Ed(k), \quad (1)$$

where $x(k) \in \mathbb{R}^n$ is the state, $u(k) \in \mathbb{R}^p$ is the input and $d(k) \in \mathbb{R}^q$ is a bounded, additive disturbance term. The index k refers to the current time instant. The system is subject to safety constraints

$$u \in \mathbb{U}, \quad (2a)$$

$$x \in \mathbb{X}. \quad (2b)$$

The nominal linear MPC controller is defined via the iterative solution of the following optimization problem

$$\min_{\bar{u}} \sum_{i=0}^{N-1} (x_i^T Q x_i + u_i^T R u_i) + x_N^T P x_N \quad (3a)$$

$$\text{s.t. } x_0 = x(k) \quad (3b)$$

$$x_{i+1} = Ax_i + Bu_i, \quad \forall i \in [0, N-1] \quad (3c)$$

$$x_i \in \mathbb{X}, \quad \forall i \in [0, N-1] \quad (3d)$$

$$u_i \in \mathbb{U}, \quad \forall i \in [0, N-1] \quad (3e)$$

$$x_N \in \mathbb{X}_T \quad (3f)$$

where x_i and u_i denote the predicted state and input at the discretization point i of the optimal control problem and the matrices Q , R and P are weighting matrices of appropriate dimensions. Note that the index i is used for the dynamics on the prediction horizon, while k is the actual time index. Stability and recursive feasibility can be ensured via the choice of the terminal set $\mathbb{X}_T$ and the terminal cost $x_N^T P x_N$. We use the notation $\bar{u}$ whenever we refer to the vector formed by the candidate input trajectory u_0 to u_{N-1} . The control law itself is given via $u(x(k)) = u_0$.

It is well known [25] that the application of this nominal MPC scheme (see [21] for a thorough discussion of constrained nominal MPC) does not guarantee recursive feasibility of the optimization problem in the presence of the unknown disturbance $d(k)$. This can be prevented by the application of Tube-MPC, where the problem (3) is modified to ensure that the constraints are satisfied robustly. The concept utilized in this work is based upon the work of [24] and will only be introduced briefly here for the sake of brevity. First, a pre-stabilizing feedback law $u = -Kx$ is introduced which is then used to predict the reachable sets $\mathbb{X}_{R,i}$ at the discretization points i for the resulting closed-loop system with respect to the external input $d(k)$. These reachable sets are now utilized to calculate tightened state and input constraints $\bar{\mathbb{X}}_i \subset \mathbb{X}$ and $\bar{\mathbb{U}}_i \subset \mathbb{U}$ by applying the pontryagin set difference (see [24]), resulting in the new optimization problem

$$\min_{\bar{u}} \sum_{i=0}^{N-1} (x_i^T Q x_i + u_i^T R u_i) + x_N^T P x_N \quad (4a)$$

$$\text{s.t. } x_0 = x(k) \quad (4b)$$

$$x_{i+1} = Ax_i + Bu_i, \quad \forall i \in [0, N-1] \quad (4c)$$

$$x_i \in \bar{\mathbb{X}}_i, \quad \forall i \in [0, N-1] \quad (4d)$$

$$u_i \in \bar{\mathbb{U}}_i, \quad \forall i \in [0, N-1] \quad (4e)$$

$$x_N \in \bar{\mathbb{X}}_T \quad (4f)$$

The implementation of this tightening will be done via calculation of the set dynamics based on ellipsoids [36] for the linearized system dynamics. Note that this approach can be further justified by the successful application of controllers obtained from the linearized dynamics for real-world autonomous racing [5]. The reachable sets around the nominal trajectory p are described via ellipsoids of the form

$$E(p, M) := \{x \in \mathbb{R}^n | (x - p)^T M^{-1} (x - p) \leq 1\}, \quad (5)$$

with p being the center of the ellipsoid and M the corresponding shape matrix. An over-approximation of the Minkowski sum of two ellipsoids [37] is given by

$$\begin{aligned} & E(p_1, M_1) \oplus E(p_2, M_2) \\ & \subset E(p_1 + p_2, (1 + c^{-1}) M_1 + (1 + c) M_2), \end{aligned} \quad (6)$$

with $c = \sqrt{\text{Tr}(M_1)/\text{Tr}(M_2)}$ and $\text{Tr}(M)$ the trace of the Matrix M . Additionally, the following definition for the affine dynamics is used

$$A \cdot E(p, M) + b = E(Ap + b, AMA^T). \quad (7)$$

IV. APPLICATION TO AUTONOMOUS VEHICLE RACING

A. Controller Concept

The results obtained under real-world racing conditions for a point-mass-based control design in [5], [11] demonstrate that simple dynamic models enable sufficiently accurate control as long as the tire constraints are considered in the form of suitable acceleration constraints during the planning process. This idea

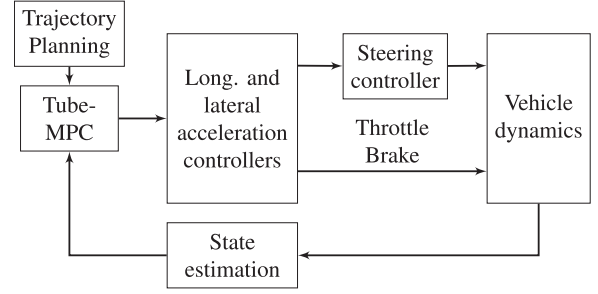


Fig. 3. Block diagram of the proposed Tube-MPC algorithm.

has been leveraged for the design of a Tube-MPC control concept in [32] and has shown promising results for a classical trajectory tracking formulation. The following section will extend the controller presented in [32] by a detailed discussion of the modifications and extensions required to design a fully functional control system under real-world conditions at one of the fastest oval racetracks in the world.

The fundamental philosophy of the concept presented is a strict split into certain and uncertain dynamics. We consider the relation between the accelerations in the vehicle frame and the actual movement to be exact, as they are based on theoretically sound differential equations without parametric uncertainties. This setting is well-suited for handling the dynamics and acceleration constraints in an MPC scheme. In contrast, the effect of throttle, brake pressure, and steering requests on the actual accelerations is subject to uncertain dynamics (such as turbocharger effects, neglected actuator dynamics, etc.) as well as parametric uncertainties (steady-state errors of the steering actuator due to load on the tires, engine calibration changes compared to the setup measured on the dynamometer, etc.). We handle this situation by introducing low-level PID-feedback loops for the longitudinal and lateral acceleration and a steering feedback loop. This design is favorable as it reduces the remaining uncertainties via fast and simple-to-tune control loops without increasing the complexity of the prediction model in the MPC. The remaining uncertainties and tracking errors are assumed to be external disturbances and considered by extension of the MPC scheme to a Tube-MPC scheme. The resulting controller architecture is depicted in Fig. 3.

The interfaces to the software stack are the trajectory planning and the state estimation. The former provides a path with continuous position and heading and a continuous velocity. The curvature and longitudinal acceleration do not have to be continuous, this is an advantage of the applied reoptimization concept (see Sec. III-A). The state estimation provides a localization in global coordinates and velocity and acceleration estimates. This component will be introduced briefly in Sec. IV-E and is based on the work done in [38].

B. Formulation of the Optimization Problem

The upcoming section summarizes the detailed derivation of the vehicle dynamics and its constraints presented in [32]. Using a model derivation procedure in curvilinear coordinates as in [6], [19] (also see Fig. 4) and a point-mass model to describe the

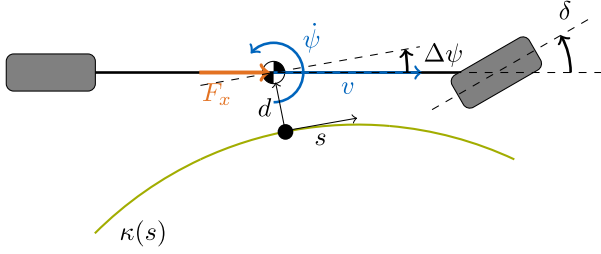


Fig. 4. Visualization of the curvilinear coordinate system and the dynamic model used to describe the vehicle behavior.

behavior of the vehicle [11] we can write

$$\begin{bmatrix} \dot{s} \\ \dot{v} \\ \dot{d} \\ \dot{\Delta\psi} \end{bmatrix} = \begin{bmatrix} (1 - d\kappa(s))^{-1} v \cos(\Delta\psi) \\ \frac{F_x}{m} - \frac{c_w}{m} v^2 \\ \dot{d} \\ \left(\frac{F_x}{m} - \frac{c_w}{m} v^2\right) \sin(\Delta\psi) + \cos(\Delta\psi) (a_y - \kappa(s) \dot{s}v) \end{bmatrix}, \quad (8)$$

with

$$\Delta\psi = \arcsin\left(\frac{\dot{d}}{v}\right). \quad (9)$$

The state s represents the progress along the reference path with curvature $\kappa(s)$, while d depicts the lateral deviation and $\Delta\psi$ the orientation error of the chassis towards this path. We use the coordinate transformation $\dot{d} = v \sin(\Delta\psi)$ to replace the orientation error with the derivative of the lateral error $\dot{d}$. The longitudinal dynamics are completed via the vehicle speed v , the longitudinal force input F_x , the vehicle mass m and the lumped drag coefficient c_w . This formulation emphasizes the interpretation of the longitudinal and lateral accelerations as the key variables to influence the motion of the vehicle and will be useful during the efficient realization of the embedded optimization algorithm.

Investigations with high speed autonomous racecars have shown that circular limits for the tire forces based on Kamm's circle do not necessarily transform directly into reliable acceleration limits on the vehicle level [39]. The actual shape of the vehicle acceleration limits depends on several factors, such as the tire characteristics and the mechanical vehicle setup. We use a diamond-shaped constrained set

$$-1 \leq \frac{a_x}{\bar{a}_x} + \frac{a_y}{\bar{a}_y} \leq 1 \quad (10a)$$

$$-1 \leq \frac{a_x}{\bar{a}_x} - \frac{a_y}{\bar{a}_y} \leq 1 \quad (10b)$$

with the acceleration limits $\bar{a}_x$ and $\bar{a}_y$ in the following as it allows to exploit the pure lateral and longitudinal vehicle capability while being cautious in the combined cases. This behavior delivered a good compromise between performance and reliability, while being easy to integrate in numerical algorithms due to its polytopic form. The limits will be obtained based on the speed of the target trajectory for each discretization point, similar as the curvature of the path, from a look-up table within the trajectory planner. This table is generated based on extensive simulations

as well as real-world vehicle tests. In addition to the acceleration constraints, the trajectory planner generates the lateral deviation limits dynamically from the situation. They are implemented via simple box constraints and can be different between the left and the right side of the driving tube.

$$d^- \leq d \leq d^+. \quad (11)$$

The actual MPC problem is formulated with respect to the target velocity profile v_R using the state transformation $\Delta v = v - v_R$ and the input transformations

$$F_x = c_w v^2 + m \frac{v \cos(\psi_P)}{1 - d\kappa(s)} \frac{\partial v_R(s)}{\partial s} + m \Delta a_x, \quad (12a)$$

$$a_y = \kappa(s) \frac{v^2 \cos(\psi_P)}{1 - d\kappa(s)} + \frac{1}{\cos(\psi_P)} (\Delta a_y - a_x \sin(\psi_P)). \quad (12b)$$

In addition, we assume the progress along the reference line $\dot{s}$ and curvature values $\kappa(s)$ to be known a-priori to the optimization problem and can therefore drop the state s from the state vector and simplify the optimization problem to use the following system dynamics:

$$\begin{bmatrix} \Delta \dot{v} \\ \dot{d} \\ \dot{\Delta\psi} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \Delta v \\ d \\ \Delta\psi \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \Delta a_x \\ \Delta a_y \end{bmatrix}. \quad (13)$$

This linear dynamic system allows us to formulate the robust MPC problem in the form of (4). We add the constraints (10) and (11) via linearization of (12). The uncertainty tube sets for the constraint tightening are calculated using ellipsoids and a pre-stabilizing controller based on a reasonable LQR weight design [5], [32]. We assume the uncertainties to enter the dynamics as matched disturbances, i.e., via acceleration disturbances acting the same way as Δa_x and Δa_y . They will serve as a tuning parameter to blend between an aggressive driving style (small uncertainties) and a rather cautious behavior (large disturbances) and are gain-scheduled based on the vehicle speed. The linearization is done based on a velocity profile obtained from a weighted linear combination of the last steps solution and the target velocity profile. Incorporating the latter allows the MPC to adjust quickly to a change in target trajectory that frequently occurs in multi-vehicle racing scenarios.

The system is discretized using Euler forward discretization and a sample rate of 60 ms. The horizon is chosen to be 40 steps which leads to an overall prediction time of 2.4 s. The QP is implemented using OSQP [40], an alternating direction method of multipliers (ADMM)-based first-order solver. The highly constrained nature of the optimization problem suits the properties of the solver using a projection approach to handle constraint satisfaction [39]. Its efficient implementation allows to execute it at an update rate of 100 Hz.

The cost function design is based on the choice of the weighting matrices Q and R . We tune the weights for the longitudinal dynamics to reflect a very gentle tracking behavior with low convergence rates to enable smooth vehicle inputs while still keeping tracking of the target velocity profile. This is important to handle situations where two vehicles closely follow each

other. The lateral dynamics weights are tuned to minimize the lateral motion in the driving tube and therefore only consider the lateral deviation very briefly. In addition, we introduce a regularization term penalizing the acceleration differences between $a_{x,i}$ and $a_{x,i+1}$ as well as $a_{y,i}$ and $a_{y,i+1}$ between the discretization points to further smooth the vehicle behavior.

C. Low Level Controllers

The lateral and longitudinal acceleration controllers serve an important task in the proposed control concept: They ensure that the assumptions used by the Tube-MPC are met by the vehicle dynamics, even in the presence of model uncertainty and external disturbances.

The lateral acceleration controller consists of three main parts: a feedforward element based on the kinematic relations of the vehicle, a proportional feedback, and an integral feedback part. The resulting control equation can be summed up to

$$\delta = \frac{1}{v^2} (a_y l + K_{p,y} \tilde{a}_y + \delta_i), \quad (14a)$$

$$\dot{\delta}_i = K_{i,y} \tilde{a}_y - \left(\frac{\delta_i}{c_1 \tilde{a}_y} \right)^3 \quad (14b)$$

with the controller gains $K_{p,y}$ and $K_{i,y}$ and the lateral acceleration control error $\tilde{a}_y$. The second element in the integrator equation above serves as a soft limiter for the integral controller part. It prevents the controller from winding up in case of significant understeer, which would lead to degraded control performance on turn out. This limit depends on the absolute acceleration limit $\tilde{a}_y$ as well as a tuning factor c_1 to adjust the margin where the integral part can compensate for deviation from a neutral vehicle behavior. The above equations are slightly modified for low-speed scenarios (close to walking speed) to limit the resulting gain and prevent oscillations via the introduction of a lower speed limit for the evaluation of the equations. The steering controller is built based on a classical PI-controller with anti-windup mechanisms. The details are omitted for the sake of brevity. It allows to shape the behavior of the steering actuator freely as the internal control loops are not exposed on the available interface.

The longitudinal acceleration controller is built slightly simpler: a feedforward element based on the vehicle mass and drag resistance and a proportional feedback term. This removes the necessity for an anti-windup strategy and simplifies the tuning procedure. The control equation can be written as

$$F = ma_x + c_w v^2 + K_{p,y} \tilde{a}_x \quad (15)$$

with the control gain $K_{p,y}$, the longitudinal acceleration control error $\tilde{a}_x$ and the lumped drag coefficient c_w . The resulting force is converted to a corresponding throttle position or brake pressure requests by using empirical lookup tables.

D. Banking Influence

One of the challenges for vehicle dynamics control is the influence of the track banking on the required steering angle and the measured accelerations. While this has been handled

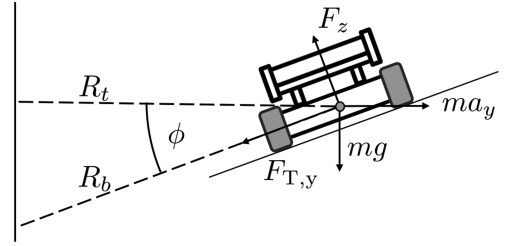


Fig. 5. Forces acting upon the vehicle in a banked turn, with R_b being the banked turn radius and R_t being the radius in the 2D-plane.

in optimal control problems for laptime minimization several times [17], it has not yet been considered during real-time-critical implementations. As this would complicate the resulting optimal control problem significantly, we propose to handle these effects on the level of the acceleration controllers. To calculate the required transformation we use the relation of the acting forces from Fig. 5. Using the balance of forces in each of the two dimensions, we can write

$$\cos(\phi) F_{T,y} + \sin(\phi) F_z = ma_y, \quad (16a)$$

$$-\sin(\phi) F_{T,y} + \cos(\phi) F_z = mg, \quad (16b)$$

with ϕ being the banking angle. The IMU which is used for lateral acceleration feedback measures the specific force acting in locally upon the vehicle which can be calculated as $a_{IMU} = \frac{F_{T,y}}{m}$. Rearranging the above equation, we can calculate the corresponding effective acceleration in the level plane to be

$$a_y = a_{IMU} (\cos(\phi) + \tan(\phi) \sin(\phi)) + \tan(\phi) g. \quad (17)$$

In addition, we have to modify the feedforward portion of the lateral acceleration controller to account for the increased turn radius R_b in the banked plane compared to the 2D track plane R_t . This can be done by modifying the first term of (14) to be $a_y l \cos(\phi)$ instead of $a_y l$.

E. State Estimation

The proposed Tube-MPC requires the determination of the lateral error, the lateral error derivative and the velocity error. All of these quantities are based upon a reliable estimation of the vehicle position and motion state in the euclidean frame. To achieve this, we use a Kalman-Filter-based concept with similar model assumptions as the proposed Tube-MPC. Details on this can be found in [38]. However, as this approach is based on a two-dimensional vehicle model and relies heavily on the IMU signals, the banking has a significant influence on the estimation quality. The transformations presented above are therefore also applied to the inputs of this estimator to compensate for these effects.

The lateral error is calculated based on a matching algorithm of the current position with respect to the path in a two dimensional euclidean coordinate frame. First, the closest section of the target path is determined. Based on this, a refined estimate is calculated from a linear projection onto this path. The lateral error derivative is determined by simple numerical derivation

and low-pass filtering. A heading and side slip angle based approach as proposed by [5] did not show satisfactory results, as the side slip angle could not be estimated with sufficient quality (sub-degree accuracy) on the used vehicle platform.

V. SIMULATION RESULTS

A. Simulation Setup

The following section evaluates the performance of the presented algorithm in comparison to an LQR controller [5] and a wide-spread MPC formulation in path coordinates using a nonlinear Pacejka tire model (based on [19]). It further conducts a parameter study to clarify the sensitivity and generate guidelines for tuning the controller on the real vehicle.

The simulation environment is based on a nonlinear dual-track model with vehicle parameters aligned with an exemplary vehicle, including actuator and sensor models. The model and the TMPC implementation are available at [41]. All results in this section are generated using a simplified version of the trajectory planning algorithm and the full state estimation and control software required for running the real-world vehicle. The Tube-MPC is implemented according to the details provided in Sec. IV-B and achieves runtimes of approx. 1 ms to 4 ms on a standard Core i7 laptop with 2.7 GHz and total control software execution rate of 100 Hz. The MPC with the nonlinear tire model is implemented using the C-Code generation framework of the library *acados* [42] and achieves runtimes of approx. 20 ms to 40 ms with a discretization of 40 ms and, similar to the Tube-MPC, 2.4 s prediction horizon. To account for the larger computational burden, we run this version with an update frequency of 25 Hz.

B. Evasion Maneuver

One of the main advantages of MPC is the rigorous handling of input and state constraints in situations where feedback is necessary to ensure that the system behaves as expected. This becomes especially important for the proposed Tube-MPC concept. It does not aim at perfect lateral tracking but instead focuses on reoptimizing the target trajectory presented by the local trajectory planner. A common challenge for the proposed software architecture (see Fig. 3) is the frequent update and modification of the target trajectory generated by the local trajectory planner. Every update of the perceived opponents and their predicted behavior may lead to a behavior change in the local trajectory planning. e.g., aborting an overtaking maneuver as it might not be feasible anymore or initiating an overtaking maneuver as an obstacle was detected nearby. These scenarios are difficult to reproduce as the ego vehicle actions directly influence the behavior of other vehicles on the track during multi-vehicle simulations. We therefore propose a double-lane change maneuver to mimic the situation that a relatively slow opponent vehicle is perceived in front of the ego-vehicle. We compare two scenarios: In the first scenario, the opponent vehicle is perceived at about 50 m in front and an evasion maneuver has to be initiated immediately. While this might seem like an extreme scenario, the significant banking of the racetrack and the low height of the vehicles make

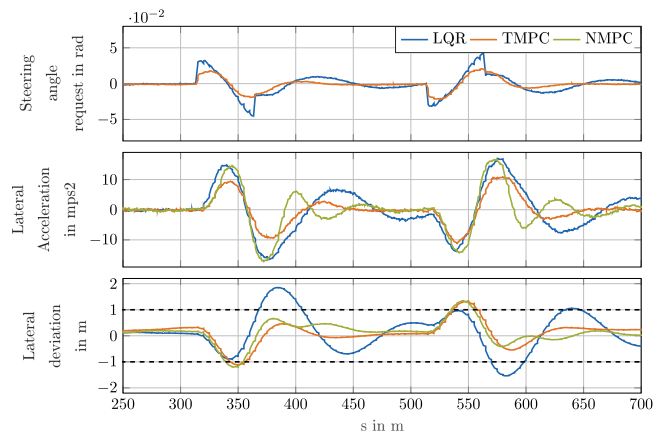


Fig. 6. Control performance during an evasive maneuver with 50 m/s with no foresight on the lane change maneuver.

this a realistic worst-case assumption. In the second scenario, the opponent vehicle is perceived at about 70 m in front, which gives the controller an additional foresight horizon to react much smoother.

The first scenario is depicted in Fig. 6 and the top plot of Fig. 8. The target trajectory switches to the lane change trajectory when the vehicle moves past the 320 m mark and it has to react immediately. The same happens at the 520 m mark. While all concepts have access to the exact curvature of the target path, the LQR controller fails to achieve the required control performance and to stay within the 1 m bounds of the target path even though it utilizes accelerations of 15 ms^{-2} . The model predictive concepts (Tube-MPC and MPC with nonlinear tire model) improve upon constraint satisfaction. However, both of them still struggle to achieve the required performance. Multiple points can be observed here: First, the non-continuous curvature profile (which reflects the reduced quality requirements for the local trajectory planner) can be handled much better by the optimization-based control methods as they generate smoother steering angle behaviors. Second, the scenario itself is challenging from a system dynamics point of view. The delays and response times can only be partially overcome when the change to the trajectory is not known with sufficient foresight. Third, the Tube-MPC shows less aggressive steering interventions for longer durations than the MPC with a nonlinear tire model. This is because the MPC with a nonlinear tire model does not have an incentive to value driving stability and can therefore create the required lateral acceleration aggressively via the front axle. In conjunction with the inherent system uncertainties, this leads to an overshoot in lateral acceleration. While this might be mitigated with increased model accuracy, cost function tuning, and higher controller update frequency, the Tube-MPC controller achieves comparable lateral control performance at a fraction of those tuning efforts. It should also be noted that the Tube-MPCs incentive (stemming from the point-mass model) to focus on steady-state cornering scenarios exploits the vehicle stability properties and therefore increases the practical applicability.

This scenario is probably the most extreme - in most cases, the local trajectory planner is well aware of what's coming

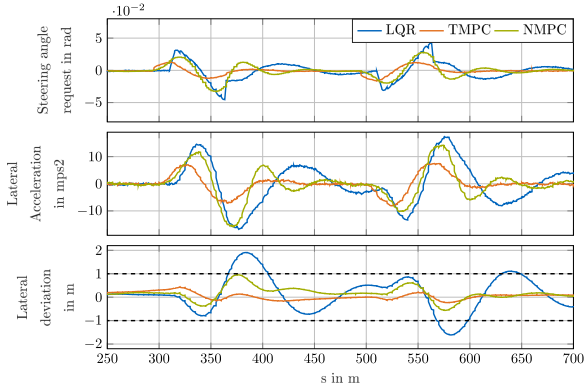


Fig. 7. Control performance during an evasive maneuver with 50 m/s with 20 m foresight (equivalent to 400 ms) on the lane change maneuver.

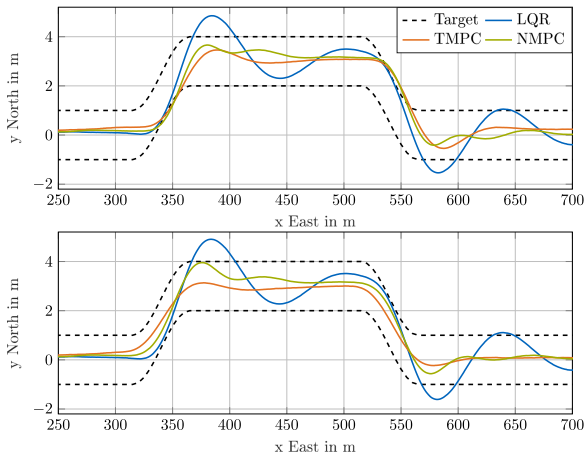


Fig. 8. Visualization of the motion control performance during an evasive maneuver with 50 m/s within the cartesian coordinate frame. The top plot shows the performance when the lane changes are not known beforehand to the software. The bottom plot depicts the improvements in performance for a foresight horizon of 20 m roughly equivalent to 400 ms.

up, even though it still changes the target trajectory within the optimization horizon. The second scenario (see Fig. 7 and the bottom plot of Fig. 8) with an approximate foresight of 20 m (roughly equivalent to 400 ms) is used here to demonstrate the advantages of the Tube-MPC concept. In this scenario, the same target path is used, but it is activated as soon as the vehicle passes the 300 m mark (the 500 m respectively for the second lane change). This foresight allows the model predictive controllers to start the lane change much more gently and therefore reduces the maximum lateral acceleration by approx. 25 % without sacrificing control performance. Again, the focus on steady-state cornering situations of the Tube-MPC leads to less aggressive steering interventions and smoother and more robust handling of the situation.

C. Importance of the Low Level Controllers

The proposed Tube-MPC concept builds around a simplified vehicle model (in contrast to other concepts in the literature, which tend to rely on more complex models for MPC [6], [31]) in conjunction with tight tracking of the target accelerations by low-level PID-like controllers. While this removes the

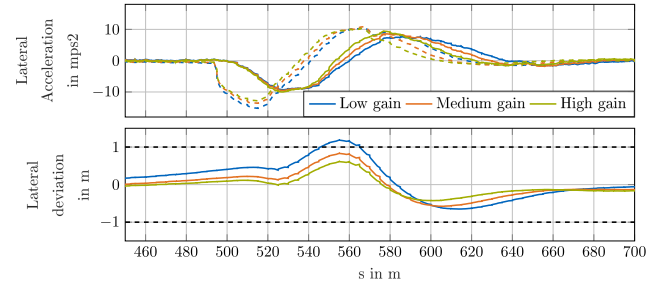


Fig. 9. Lateral acceleration controller influence during an evasive maneuver with 65 m/s. The dashed lines depict target values while the solid lines depict actual measurements.

need for sophisticated tire models, it becomes essential that the acceleration response of the vehicle matches the expectations of the model predictive controller. It should be noted that this requirement is even more pronounced as the Tube-MPC concept is heavily based on the idea of constraint satisfaction rather than target tracking. While a model/reality mismatch would only lead to slightly increased control values in a classical tracking concept, the violation of the underlying (uncertain) predictions of the vehicle motion almost certainly leads to significant violations of the state constraints and results in strong corrective actions by the Tube-MPC.

Fig. 9 depicts the influence of lateral acceleration control quality on the overall control system performance. The figure shows three different tunings from the low-level acceleration controller, ranging from a rather gentle tuning to a high-gain setup. The high-gain setup keeps the vehicle in the allowed driving tube with moderate corrections and shows better acceleration tracking. The comparison between the target accelerations calculated by the Tube-MPC controller between 500 m and 520 m shows significantly increased target accelerations for the gentle feedback setup (15 ms^{-2} instead of 12 ms^{-2}). This behavior can be explained by the fact that the vehicle approaches the lateral deviation constraint and violates it, which asks for strong corrective actions from the Tube-MPC. It should be noted that this is only possible as long as the acceleration constraints permit this increased feedback and therefore the low-level tracking becomes even more vital when the acceleration constraints are approached.

D. Influence of the Optimization Horizon

While the number of optimization variables is limited by the resulting computational load, the length of the optimization horizon can be adjusted independently from this via the discretization time. We run the optimization with an update rate of 100 Hz and vary the optimization horizon between 2.0 s and 2.8 s. The results of this experiment are depicted in Fig. 10. As expected from a theoretical perspective, longer horizons lead to improved control quality via a decrease of the required lateral accelerations to move through the double lane change maneuver. While the lateral acceleration peaks at 10 ms^{-2} for the 2.0 s horizon, this is reduced to 8 ms^{-2} for the 2.8 s horizon. This improvement of 20% is significant when driving at the handling limits of the vehicle. It should be kept in mind that these

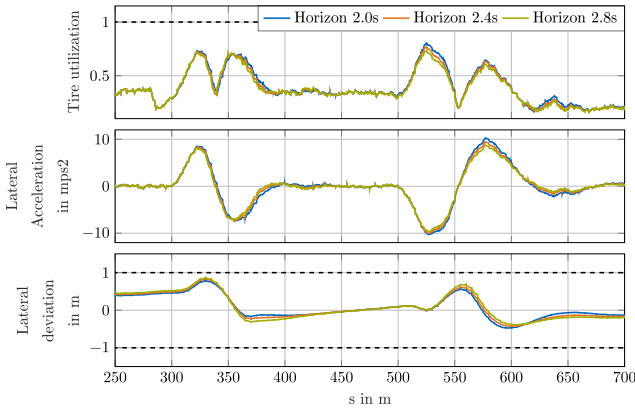


Fig. 10. Influence of the optimization horizon during an evasive maneuver with 65 m/s.

improvements stay only valid if the target trajectory does not change while the vehicle is driving the maneuver (see previous sections for a discussion of this issue) and longer horizons increase the computation time of the local trajectory planner. We choose 2.4 s to be a viable compromise between those targets as it still allows to cover 180 m of foresight horizon at top speed (approx. 75 m/s)

VI. EXPERIMENTAL RESULTS

A. Vehicle Platform

The autonomous race vehicle used is a modified Dallara Indy Lights vehicle, the Dallara AV-21. A drive-by-wire system actuates its steering and brakes, and its powertrain and gearbox can directly be actuated via the engine control ECU. The vehicle is equipped with two independent GPS units by Novatel, three LIDAR units by Luminar, three radar sensors by Aptiv, and six cameras by Allied Vision. In addition, standard vehicle dynamics sensors such as engine speed, wheel speeds, and brake pressures are available. The autonomous driving software runs on an ADLink x64 computer system based on an Intel Xeon with eight physical CPU cores and an NVIDIA RTX 8000 GPU.

The proposed controller requires strict real-time capability of the underlying computer platform. To enable this on the Ubuntu 20.04 based operating system, we allocate a dedicated CPU core, set appropriate scheduling properties, and move all file system interactions (e.g. data logging) from the vehicle motion control software to external, potentially asynchronous, software components. The rest of the software stack is interfaced via the standard robotics middleware ROS2 Galactic with Cyclone DDS as a transportation layer.

B. Single Vehicle Qualifying Competition

The following experiments have been conducted on the Las Vegas Motor Speedway Oval circuit. The track is 2440 m long and has a maximum banking of 20°. The results of the single-vehicle qualifying run of the passing competition on January 7th are depicted in Fig. 11. It covered various speeds between 45 m/s and 70 m/s, with two of those laps shown here. Interestingly, the

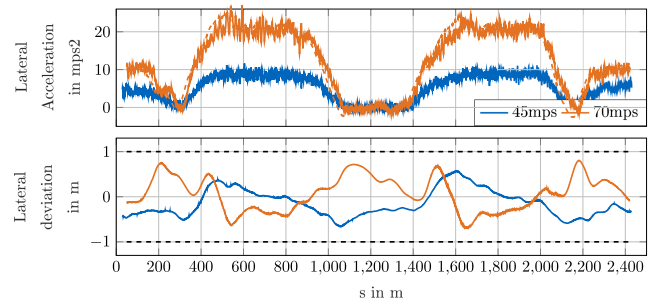


Fig. 11. Controller performance on the real world vehicle at various speeds. The dashed lines indicate target accelerations, while the solid ones indicate actual accelerations (transformed into the 2D plane).

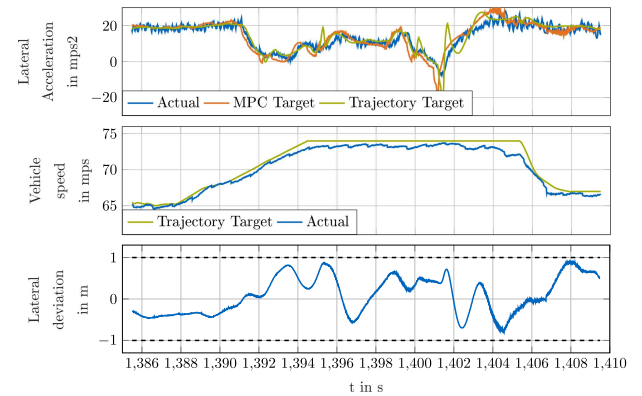


Fig. 12. Controller performance on the real world vehicle during the final competition run overtakes between PoliMOVE and TUM Autonomous Motorsport. Accelerations have been transformed into the 2D plane.

lateral movement does not increase significantly with increasing lateral accelerations as it would be expected as a response to increasing uncertainty. In contrast, it can be seen that the controller leverages the available lateral movement to reduce the required control action to compensate for inaccuracies due to the nonlinearities in the model (see 500 m and 1600 m). In classical tracking formulations, the controller would have added additional lateral acceleration when the vehicle is approaching the target of zero quickly. In the proposed concept, it allows to actually move beyond this target and starts to react only when it gets close to the actual constraint. This results in less aggressive steering corrections and therefore higher driving stability. Overall, this behavior is much closer to how a real driver would drive those turns. They focus on the overall task rather than strictly tracking artificial target trajectories.

C. Two Vehicle Passing Competition

In addition to the single-vehicle runs, the concept was proven during several experiments with two vehicles. The final competition run on January 7th has been conducted by the PoliMOVE and the TUM Autonomous Motorsport team (see Fig. 12 and [43] for a video) and includes multiple overtakes at varying speeds between 45 m/s and 74 m/s. The acceleration plot shows that the trajectory planning does not consider jerk constraints or costs and is therefore prone to high gradients in lateral

and longitudinal acceleration (Trajectory Target). The proposed Tube-MPC controller reduces those significantly (MPC Target in contrast to Trajectory Target), which results in a smooth vehicle behavior (Actual).

VII. CONCLUSION AND FUTURE WORK

We presented a Tube-MPC concept for operating a full-scale autonomous racecar at speeds up to 265 kmh^{-1} and lateral accelerations up to 21 ms^{-2} in single and two-vehicle scenarios. In contrast to classical tracking control concepts, we leverage the advantages of optimization-based control to allow coarse target trajectories and, therefore, increase the software stack's overall performance. This is made possible by explicitly considering the uncertainties within the prediction model, which generates a behavior that is cautious for long-term predictions but exploits the full vehicle potential on near-term predictions.

A notable feature of this strategy is that it lumps all uncertainties within a few uncertainty parameters and it's therefore sufficient to use a friction-limited point-mass model as dynamic model. This is especially helpful in the face of the significant banking influence on oval racetracks, as this is harder to handle appropriately in more sophisticated models with nonlinear tire and suspension effects. In addition, this strategy renders the concept independent from potential setup or parameter changes, e.g., a balance shift to make the vehicle more understeering.

While this paper demonstrated the huge potential of relatively simple dynamic models when combined with appropriate low-level controllers and thorough uncertainty handling, reaching super-human performance will likely require more advanced control strategies. An important improvement area is the controller's stabilization capabilities even when far away from the stable operating points of the state space (e.g., while drifting with large chassis side slip angles), as the proposed control concept relies heavily on the inherent stability of the vehicle. Even though this could be handled via more complex models [15], further work to include robustness properties and make it real-time capable has to be done to apply nonlinear tire models within an NMPC scheme for autonomous racing. Another promising direction is to leverage data-driven methods [31], [44] to refine the uncertain parts of the model while driving and exploit them more aggressively as soon as the models have converged. However, these algorithms remain challenging to implement in a real-time-capable manner and deploy to work reasonably well along all operating conditions of the vehicle.

CONTRIBUTIONS AND ACKNOWLEDGMENTS

Alexander Wischnewski developed, analyzed and implemented the proposed control concept. Thomas Herrmann and Frederik Werner contributed to the real-time capable implementation and integration in the software stack of the TUM Autonomous Motorsport team. In addition, all of them have been involved in running the vehicle safely on multiple racetracks and finding the optimal setup for the competition runs. Boris Lohmann contributed to the critical revision of the manuscript and the conception of the research project.

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autonomous software stack modules, and evaluation of overall vehicle performance.



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